

Cancer Pharmacogenomics Demo Documentation

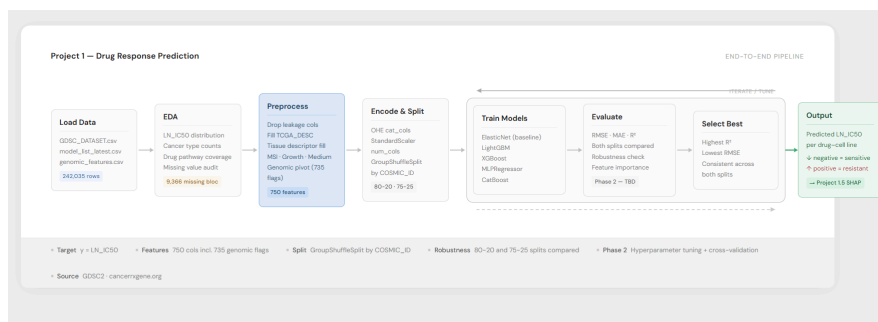


Figure 1: In the above figure, there is a pipeline of the plan we made for this project. As it shows, we are wrapping up the preprocessing phase and moving forward to encoding and splitting the data.

Current Overall Progression

Currently, we have finalized the data preprocessing, cleansed the GDSC datasets along with the new support dataset, and prepared the data for modeling by handling missing values. The data sets were merged and the categorical features were encoded. We started to play around with different options for the baseline model. In Figure 1 above, it shows the progress we have made and is a visualization of what we still have to complete. Since the preprocessing stage has ended we are moving on to encoding and splitting the data categorizing the important columns we would like to use.

Dataset/EDA

The main dataset used for this project is obtained from the Sanger Institute experiments, and later, supplementary CSVs from the same source were found that helped with the preprocessing efforts. The main GDSC dataset includes key features like LN_IC50 (molecular target of the drugs) as the target variable, TCGA (description of the cancer type) labels, and the drug names, but unfortunately the dataset came with several missing values and label inconsistencies that needed to be addressed before modelling. The Cell_Lines_Details.csv included information on tissue descriptors, growth properties, and other key features, but some fields were missing. The genomic_features.csv would not only help fill in those missing values but also would add entirely new features like gene mutations and CNAs that gave the model a stronger biological signal.

The EDA was primarily done on the main dataset before the supplementary CSVs were found. This EDA of the first dataset eventually helped with realizing many things about the data, leading to changes in the way the group handled the dataset for the models and predictions to be done. The first EDA was done on the LN_IC50 (primary variable) column to check its distribution and missing values. It had a wide range of about -8.75 to 13.82 , with a heavy tail, meaning there were many extreme sensitivity/resistance values, though it presented no missing data that needed to be fixed. Missing values were checked for in other columns, which led to the finding of columns with missing values. The remaining columns were then checked for missing values, and found that the "PUTATIVE_TARGET" column had the most notable issues. Additional columns from other parts of the dataset were identified, which could potentially improve the prediction accuracy. This process led to the two approaches for the rest of the project: either dropping or filling the rows with missing values.

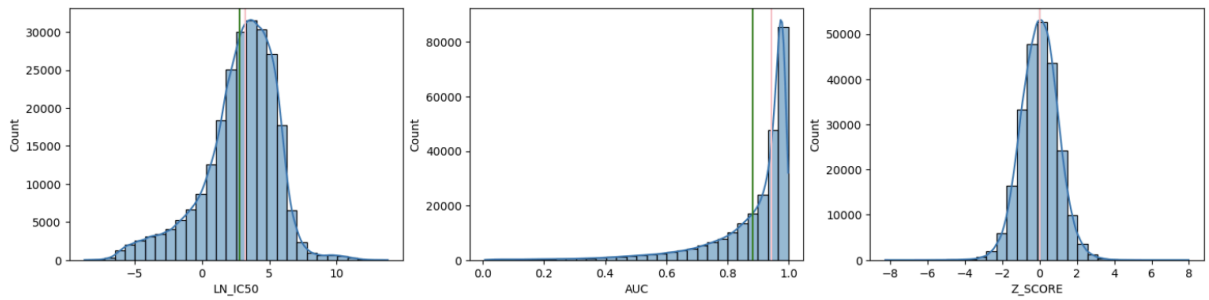


Fig 2- The 3 Numerical Features including LN_IC50, AUC and Z_Score (Pink is median, green is mean)

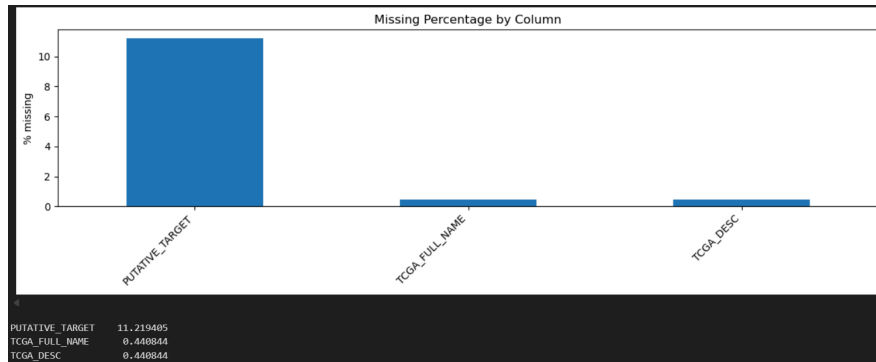


Figure 3 - Missing values Bar chart for the 3 columns

https://github.com/Temple-University-CIS-4496/cis-4496-project-cancer-pharmacogenomics/blob/master/Code/Allan%20Thomas/Project_EDA.ipynb

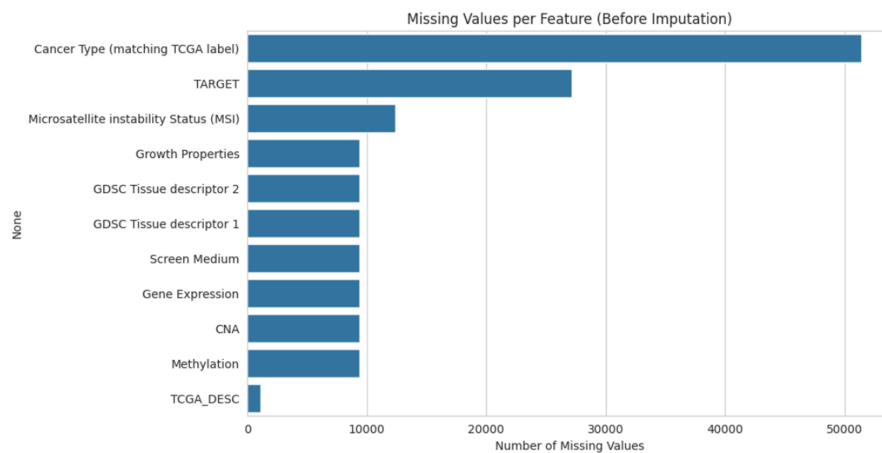


Figure 4 - Missing Values per Feature

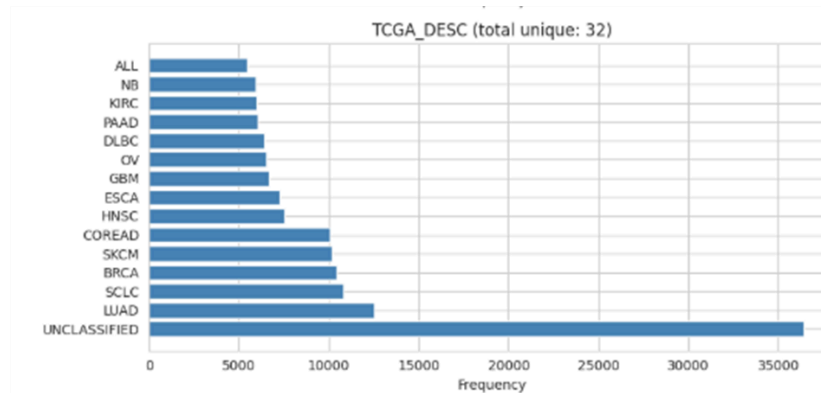


Figure 5- Visualization of the TCGA_DESC feature

Preprocessing

The preprocessing was done individually for everyone in the group to have the opportunity to learn the data. There were multiple different ways to preprocess. One approach was to load the GDSC dataset and remove rows with missing or UNCLASSIFIED cancer types as well as dropping identifier columns that carry no predictive value. Categorical features were then encoded into numeric values using either label encoding or one-hot encoding. Another method that was applied was the StandardScaler method to numerical features. Through EDA, it was agreed on to flag and drop the UNCLASSIFIED category in TCGA_DESC as it did not represent a real biological classification. An 80/20 train/test split was completed by a member of the group which produced a final dataset of approximately 195,000 rows ready for modelling. There are still a few more issues with merging and removing unwanted rows with UNCLASSIFIED or similar variables.

The second approach was instead of dropping rows and missing values, we decided to fill them using supplementary CSVs from the same institute (the Sanger Institute) that provided the main dataset. The cross-filled data rows including missing tissue descriptors, growth properties, and MSI status were cross-filled using the Cell_Lines.csv, while using the Genomic_Features.csv. These were merged to recover missing values and add 735 new gene mutations and CNA columns as additional features for the model.

Preprocessing Output:

```
df shape : (242835, 19)
df2 shape: (242836, 19)

clean df
dropping null TCGA_DESC: (240968, 19)

clean df2
dropping null TCGA_DESC: (240969, 19)

merging df + df2
Merged shape: (240968, 36)
['COSMIC_ID', 'CELL_LINE_NAME_df', 'TCGA_DESC_df', 'DRUG_ID', 'DRUG_NAME_df', 'LN_IC50_df', 'AUC_df', 'Z_SCORE_df', 'GOS
C Tissue descriptor 1', 'GOS Tissue descriptor 2', 'Cancer Type (matching TCGA Label)', 'Microsatellite instability sta
tus (MSI)', 'Screen Medium', 'Growth Properties', 'CNA', 'Gene Expression', 'Methylation', 'TARGET', 'TARGET_PATHWAY', '
DATASET', 'NUPR_RESULT_ID', 'NUPR_CURVE_ID', 'CELL_LINE_NAME_df2', 'SANGER_MODEL_ID', 'TCGA_DESC_df2']

Target column used: LN_IC50_df

Merged clean shape (after dropping missing features/target): (240968, 36)
Encoded X shape: (240968, 2)
X shape: (240968, 2)
Y shape: (240968, 1)
```

Model

A few models were tested on their default statin without fine tuning parameters. It was learned that ridge, linear regression were not options to use for this dataset based on their RMSE and R-squared evaluation metric scores. Due to the long run times of Random Forest and Logistic regression models, it was ultimately decided to drop the model to conserve memory and time. Then, through testing out different models, it was found that ElasticNet seems promising for a baseline model as well as

LightGBM, CatBoost, XGBoost, along with others could be used for a primary or stronger model without any fine-tuning of the model's parameters. These models were tested with the two different methods of preprocessing spoken about above.

Model output (after dropping):

LightGBM MAE : 1.1250 R ² : 0.7221 CatBoost RMSE : 1.4779 MAE : 1.1384 R ² : 0.7177	<pre> Features: ['IC50_DESC', 'Drug_ID', 'TARGET_ID'] Train: 80000 Test: 47970 Linear Regression RMSE : 1.3914 MAE : 1.0421 R² : 0.7498 Ridge RMSE : 1.3914 MAE : 1.0423 R² : 0.7498 ElasticNet RMSE : 2.7339 MAE : 2.0828 R² : 0.0340 HistGradientBoosting RMSE : 1.4669 MAE : 1.1253 R² : 0.7219 </pre>
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Model output (after filling) -

Linear Regression (baseline) RMSE : 8.4591 MAE : 5.6838 R ² : -8.3462 Ridge Regression (baseline) RMSE : 7.1777 MAE : 4.7844 R ² : -5.7291 ElasticNet (baseline) RMSE : 2.0368 MAE : 1.6327 R ² : 0.4581	LightGBM (strong) RMSE : 1.3265 MAE : 1.0013 R ² : 0.7702 CatBoost (primary) RMSE : 1.4888 MAE : 1.1343 R ² : 0.7105
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Expected Output

After satisfactorily merging the additional information and cleaning the matrix of approximately 195,000 rows across 8 encoded features, paired with LN_IC50 as the prediction target, and split into training and testing sets. A few options are currently being evaluated, as stated above, for the baseline model and what best suits the regression problem. The ideal output from the best performing baseline model would be a close predicted LN_IC50 value per cell line and drug pair, with an R² of around 0.3–0.5 indicating the model is capturing some meaningful signal before we improve on it with stronger models. Of the preliminary models tested, ElasticNet was the strongest, with an RMSE of 2.0368, MAE of 1.6327, and R² of 0.4581. Therefore, the output will be the analysis of these results, along with further cleaning of the datasets, leading to the answer to the core question: Can we predict how sensitive a cancer cell line will be to a drug, given its biological and pharmacological profile?

Difficulties/challenges

The difficulties faced throughout the start of the project was learning and understanding the data completely. This required taking time to understand the different cell lines and how we could handle them through preprocessing. A struggle that was run into was merging the data sets together individually by a couple of group members in order to find a correlation between the missing values among the data. The one dataset, GDSC_DATA.csv, contains all the data merged into one which will be used as the primary set. There was also a struggle with handling missing values leading to two different solutions, either completely eliminating them or using supplementary data and adding it to the dataset.

Next steps

Since much of the necessary preprocessing has been completed, the next step is establishing the baseline model. The dataset is a supervised regression task with a continuous log-scaled LN_IC50 (drug response) as the target output. Therefore, ElasticNet, linear, or ridge regression makes the most sense to establish a basic model that sets the bar for baseline performance. An 80-20 and a 75-25 train/test split will be run to compare and evaluate using RMSE, MAE, and R^2 to check the model sensitivity. One of the models will be chosen by comparing better metrics and then move on to Random Forest or XGBoost as potential stronger models. Once a satisfactory baseline model with a high enough R^2 is and low enough RMSE is settled on more possible models will be explored.